

Safari Connect Bookings Analysis

SQL analysis & Power BI dashboard summary — route performance, driver performance, and revenue trends

Project Overview

This project analyzes bookings data from Safari Connect, a matatu / bus / minibus transport booking platform. SQL queries were written against a cleaned view (`safari_connect.v_clean_trips`) to answer three core business questions: which routes perform best, which drivers perform best, and how revenue is trending over time. The results feed a Power BI dashboard used for reporting.

Key Dashboard Metrics

Metric	Value
Total Revenue	KES 228,000 (227,810 exact)
Total Seats Booked	452
Highest Earning Route	RT001
Best Performing Vehicle Type	Matatu
Average Trip Rating	3 / 5 stars
Average Driver Rating	4 / 5 stars

What the SQL Analysis Covers

1. Route Analysis

- **Revenue by route** — grouped by `route_code`, `route_from`, `route_to` and summed `total_fare`, to find which routes earn the most.
- **Popularity by route** — summed `seats_booked` per route as a measure of demand.
- **Efficiency per seat** — `total_fare` divided by `seats_booked` per route, to find which route earns the most per seat sold rather than just in aggregate.

2. Driver Performance

- Revenue and average `driver_rating` calculated per driver.
- Cross-check between `driver_rating` and `trip_rating` (passenger satisfaction) to test whether higher-rated drivers produce higher-rated trips.

3. Revenue Trends

- Monthly revenue calculated with a `LAG()` window function to pull each month's prior-month revenue.
- Month-over-month percentage change: $(\text{current} - \text{prior}) / \text{prior} \times 100$.

- Best and worst month by total fare, using MAX/MIN via ORDER BY ... LIMIT 1.

Techniques used throughout: GROUP BY aggregation, window functions (LAG), CTEs (WITH clauses), derived ratio metrics, and ORDER BY / LIMIT for ranking.

Key Findings

Route Code	Total Revenue (KES)	Seats Booked	Trip Rating
RT001	51,600	41	★★★★■
RT002	38,250	40	★★★★■
RT003	20,700	43	★★★★■
RT004	43,200	51	★★★★■
RT005	7,620	62	★★★★■
RT009	7,300	33	★★★★■

Selected routes shown; RT006–RT008, RT010 fall between these in revenue.

- **Top route:** RT001 leads clearly on revenue (KES 51,600), ahead of RT002 and RT004.
- **Underperformers:** RT005, RT008, and RT009 all earn under KES 7,700 — a long tail of low-earning routes.
- **Vehicle type:** Matatu generates the most revenue share (38.6%), followed by Minibus (37.1%) and Bus (24.3%).
- **Top driver by revenue:** Isaac Korir. On the RT003 drill-through, Peter Ngugi is the highest-earning driver for that specific route.
- **Payment method:** M-Pesa dominates by a wide margin over Cash and Card.
- **Rating gap:** average trip rating (3★) trails average driver rating (4★) — driver rating alone does not appear to predict passenger satisfaction.
- **Monthly revenue:** volatile month to month, with a sharp dip around August and a peak near September/October — no clean linear growth trend.

Recommendations

- 1 **Double down on RT001.** Highest earner and likely highest demand — consider adding capacity (more Matatu trips) on this route.
- 2 **Review or sunset RT005 / RT008 / RT009.** Investigate whether low revenue stems from low demand, poor scheduling, or pricing before deciding to cut or restructure them.
- 3 **Expand Matatu allocation.** It's the most efficient vehicle type by revenue share — shifting some Bus/Minibus capacity toward Matatu on high-demand routes could lift revenue.
- 4 **Investigate the rating gap.** Since driver rating doesn't track with trip rating, satisfaction drivers are likely elsewhere (vehicle condition, punctuality, route congestion) — worth a follow-up query joining trip-level factors against trip_rating.
- 5 **Promote top revenue-generating drivers.** Use both revenue and rating data (e.g. Isaac Korir, Peter Ngugi) as justification.

- 6 **Push M-Pesa further.** Since it's already dominant, consider incentives (small discount/cashback) to migrate remaining Cash/Card users and reduce cash-handling overhead.
- 7 **Smooth revenue volatility.** Month-to-month swings suggest inconsistent demand — consider promotions in low months (e.g. August) to flatten the curve.